

Semantic Alignment of AI Models: Concept Collapse, Checkpoint Dynamics, and Cross-Lingual Transfer

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Language model benchmarking is a difficult task. Outcome reasoning alone does not test the model’s conceptualization of language and popular open-source benchmarks are quickly saturated or ingested as training data. It is important to test the model’s output, but augmenting these tests by characterizing semantic structure gives more insight to how models relate abstract concepts. However, the high dimensional embedding spaces are not easy to interpret. This work demonstrates how topological methods can be used to rigorously compare these spaces to low dimensional and interpretable baselines like ontologies and curated knowledge graphs. These multi-modal alignment tests make it possible to track model adaptations and test phrase understanding across multiple languages.¹

1 Introduction

1.1 Motivation

Large language models (LLMs) are difficult to evaluate rigorously. Outcome based benchmarks are a popular method for grading general competence, but the tests are not exhaustive due to their limited scale compared with the vast output space of the models. Red teaming and human evaluation help fill the gaps in these tests, but they are expensive and labor intensive. Popular models benefit from increased attention by surfacing more failure modes and edge cases, but even the attention of many users will leave vulnerabilities unfound. This problem is only exacerbated for less popular or more niche models, their outcome space remains extremely broad but there are fewer users to test the models.

This work demonstrates how a topological alignment approach effectively compares models’ semantic structure to curated baselines. Rather than being restricted by string output or multiple choice methods like many benchmarks or red-team approaches, this method leverages the models’ embedding space to test coherence. The advantage of these techniques when test sets are saturated is demonstrated in the companion paper (Ashoff & Rodu, 2026) and extended in this work. We show that the method effectively investigates training dynamics, cross-lingual alignment, and the manipulation of LLM behavior.

1.2 Existing Work

AI alignment focuses on how to ensure human preferences are learned by and evident in the output of models. Many papers have outlined the common failure modes of AI. These

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¹Implementation available at github.com/tylerashoff/persiscope and on PyPI as `persiscope`.

include unintended consequences caused by domain shifts or model extrapolation (Amodei et al., 2016), adversarial inputs (Zou, Wang, et al., 2023) from users with prompt injection (Greshake et al., 2023) and jailbreaking (Wei et al., 2023) (Liu et al., 2023), or more insidious model behavior like reward hacking (Bondarenko et al., 2025) and sleeper agent models (Hubinger et al., 2024). Many solutions are proposed to these problems, but Reinforcement Learning with Human Feedback (RLHF) (Christiano et al., 2017) has become an industry standard for model training. Other approaches like Anthropic’s Constitutional AI (Bai et al., 2022), and what they internally call the “Soul Doc”(Askell et al., 2026), try to impart values to the model with “a detailed description of [...] the kind of entity we would like Claude to be”(Anthropic, 2026a). To provide oversight of large models at a scale where RLHF may fail, “weak-to-strong generalization”(Burns et al., 2023) is employed, where a smaller model is trained to supervise a larger model by generalizing human interpretable safety instructions. However, other work also tries to understand if these setups can be trusted, or if LLMs can effectively fake alignment (Greenblatt et al., 2024).

The mechanistic interpretability field is also interested in model behavior, but focuses on finding “circuits” (Elhage et al., 2021), or subgraphs of a model’s network, that correlate with certain behaviors. This interest in circuits has driven work on monosemanticity (Bricken et al., 2023), sparse autoencoders (SAEs) (Cunningham et al., 2023), and transcoders (Ameisen et al., 2025) that try to isolate and identify features that activate for specific concepts.

Like alignment and mechanistic interpretability, semantic structure research tries to understand how to elicit certain model behaviors. But, rather than shaping it with human feedback, or looking for circuits, the work tries to understand the latent representations of these behaviors. The hope is that vectors can be found in the high-dimensional latent space that correspond to values or concepts in a model. The classic example shows how vectorization of words may be combined to morph one concept into another: “*King* – *Man* + *Woman* = *Queen*” (Mikolov et al., 2013). Finding and manipulating these vectors related to high-level concepts like “honesty”, “power-seeking”, or “happiness” is called Representation Engineering (RepE) (Zou, Phan, et al., 2023).

This informed Bidirectional Encoder Representations from Transformers (BERT) (Devlin et al., 2019) which introduced contextual embeddings and provided the structure for modern transformer models. Studying the emergent world representations (Li et al., 2023) of a model is an active field of research, including the representation of categorical concepts as polytopes (Park et al., 2024) and truth vectors (Marks & Tegmark, 2023). Work related to the Manifold Hypothesis in representation learning (Bengio et al., 2013) suggests that complex concepts may be associated with linear directions (Nanda et al., 2023) in the model’s representation space. Other work has leveraged this hypothesis to show that LLMs may have a coherent world view by demonstrating a spatio-temporal (Gurnee & Tegmark, 2023) understanding of places and historical figures.

1.3 Limitations of Benchmarking and Test Sets

Language models are often graded using benchmarks and open source human evaluation networks like the “Chatbot Arena”(Chiang et al., 2024). These methods are primarily task based; i.e., they are designed with a set of tasks each with a set of examples which are graded using various measures of correctness. Three popular examples of these benchmarks are MMLU, HELM, and BIG-Bench. These tasks often take the form of multiple-choice questions generated by experts or, in the case of MMLU, pulled directly from standardized tests. Often these models have been trained on most of the Internet, which includes published standardized tests. Even a stochastic parrot would excel at the recall-based tests that are used in the popular benchmarks.

Prediction focused tests are an important type of evaluation since, ultimately, the predicted word sequences are the product of these models. However, embedding spaces offer a richer,

numerical representation of these outputs. The embedded data reveals how a model builds relationships between data points and broader concepts, if the data can be grouped into classes.

Table 1: Popular Benchmarks

Benchmark	Focus	Description
MMLU	Standardization	Built directly from multiple-choice standardized tests, but becoming saturated
HELM	Safety	Developed around scenarios and question answering, more niche
BIG-Bench	Broad Task Set	Continually growing set of tasks designed to avoid saturation
Chatbot Arena	Human Preference	Humans pick between outputs from two LLMs for the same prompt

The Measuring Massive Multitask Language Understanding (MMLU) (Hendrycks et al., 2021) benchmark is “a massive multitask test consisting of multiple-choice questions from various branches of knowledge”. The questions, 15908 across 57 tasks, “were manually collected by graduate and undergraduate students from freely available sources online”. In spirit it is a multiple-choice standardized test for language models, in fact many of the questions come directly from standardized tests. It seems intuitive to grade language models like we grade students and it has been a standard metric, but it has become saturated. Additionally, since it is only a multiple-choice test, sensitivity analysis is limited by the number of questions and the trickiness of the answers.

The Holistic Evaluation of Language Models (HELM) (Liang et al., 2023) evaluation framework is described as a “living benchmark”. The benchmark’s core evaluation set is 16 scenarios and 7 metrics designed to evaluate model performance in multiple domains, and continues to add more multilingual and multimodal scenarios. It is impressive work covering many of the most powerful models and evaluating them across many dimensions. Their measure of Robustness is of primary interest to this work.

HELM “measure[s] the robustness of different models by evaluating them on transformations” of the data sets and “measure[s] the worst-case performance of a model across these transformations”. These transformations are broken into two types: “*invariance and equivariance*”. Invariance tests “natural and relatively mild” changes like typos while equivariance tests “semantics-altering perturbations”. However, the authors note that due to limitations in generating realistic semantic changes and specifying expected output changes for long output sequences, the use of these measures is relatively limited. The authors also point out, these metrics only capture the “*local* robustness of a model” and generating enough changes to approximate the true worst-case performance is “not feasible in this evaluation”. These issues motivate the need for a model characterization that captures global and local performance.

The Beyond the Imitation Game Benchmark (BIG-bench)(Srivastava et al., 2023) consists of a set of 204 language tasks, but is constantly growing as developers continue to add to the project and has been updated to BIG-Bench Hard (Suzgun et al., 2023) and Extra Hard (Kazemi et al., 2025) variants to avoid saturation. On the high end these tasks have on the order of 10^6 examples while most tasks have around 10^2 - 10^3 examples. BIG-Bench provides a set of available metrics for developers to choose from. Some of the metric options are presented in Table 2. Exact string matching is the most restrictive of all the metrics. Bleu

(Papineni et al., 2002) and ROUGE (Lin, 2004) allow for some flexibility in word choice and word arrangement. Bleurt (Sellam et al., 2020) uses a secondary model to grade the output of the first, both multiple-choice methods rely on the available choices and the calibration version incorporates the model’s own internal confidence. Yet none of them offer a natural model characterization measuring the sensitivity of these results relies on the scale of the test set.

Many types of benchmarks exists for different types of models. Agentic reasoning tests, like ARC-AGI (Chollet et al., 2026), may ask the system to solve ambiguous interactive puzzles for example. Despite, their complexity and novelty, these benchmarks still suffer from the same limitations. They are restricted by the variety and relevance of individual question sets. They do not directly measure how the model conceptualizes a domain.

The insufficiency of these metrics can be seen mostly clearly in the continued popularity of Chatbot Arena and its variants. Chatbot Arena shows a human evaluator the output from two models using the same prompt and asks the human to choose the better response. The cumulative result of many human evaluations is a crowd-sourced model ranking. This ranking fills the gap in other frameworks by capturing some ill-defined human alignment score, but it does not try to make measurement explicable or rigorous. If models are to be trusted in unfamiliar situations, understanding how the model conceptualizes the world is more important than if the model was right or wrong on an inexhaustive test set.

Table 2: *Sample of BIG-Bench available metrics.*

Metric	Type	Description
Bleu	Text-to-text	Weighted average of subsequence alignment between output and reference word sequences.
Bleurt	Text-to-text	Relies on another model, BERT, to simulate human judgment.
ROUGE	Text-to-text	Counts the overlapping subsequences between output and reference word sequences.
exact.str.match	Text-to-text	Binary indicating if the strings are identical.
multiple.choice grade	Multiple-choice	Observed probability of correct choice.
calibration multiple.choice brier.score	Multiple-choice	Observed probability of correct choice compared to the model’s internal confidence score.

1.3.1 Task-based Testing in the Real World

These benchmarks curate a large test set and evaluate the models’ output on correctness by combining scores on different tasks into a final grade. These evaluations fundamentally rely on an ever-growing and community-sourced test set. However, by the benchmarks’ own admission, it is not clear that generating an exhaustive test set is even possible given the enormous size of the output space of these models. Their scale is impressive, but *Webster’s Third New International Dictionary* (2002) has about 470,000 entries and on the order of 10^{16} combinations for 3-grams. Of course, many of these combinations will be nonsensical, but the input to language models is not restricted to proper grammar and most texts are longer than three words. This estimate also does not take into account multilingual models

or subtokenization of words. Additionally, prompt injection and jailbreaking can redirect prompt instructions or avoid built-in guardrails respectively, so both coherent and nonsense input can lead to undesirable output.

This overview has focused on language models, but the problems are similar to the issues models face in other domains. Although the output space is large, language is still a discrete problem limited in scope by the number of characters in a given vocabulary. Image, audio, or spectral data are continuous domains. While they may be discretized for processing, their true input and output spaces are infinite.

Frontier labs like Anthropic are also facing these challenges while evaluating models.

"For AI R&D capabilities, we found that Claude Opus 4.6 has saturated most of our automated evaluations, meaning they no longer provide useful evidence for ruling out [safety measures. ...]

We may build more tasks to test capabilities in the long-horizon regime, but evaluations of model capabilities may also need to depend more on expert judgments[.]"

– Claude Opus 4.6 System Card, Anthropic (2026b)

The paper introducing BIG-Bench starts with a prescient quote which suggests the opacity and complexity of modern models is inherent to competent learning machines.

"An important feature of a learning machine is that its teacher will often be very largely ignorant of quite what is going on inside."

– Alan Turing (1950)

In some respects the value of these metrics lean on Turing’s quote. If we are left with only the output for a model, we can only check its sample correctness. But we have access to the internal model embeddings, and while they may not provide us with a complete understanding of the system, we can still leverage the information the embeddings provide to peer inside. Even if we remain “very largely ignorant” of a model’s inner workings, we do not need to pretend to be completely ignorant.

The main contributions of this work are:

- Induced concept collapse can be tracked as increasingly aggressive malicious adaptations are applied
- Semantic structure alignment can be characterized during model training
- Multilingual comparisons can be made using a method that captures semantic meaning more naturally than existing methods
- Changes in a model’s internal semantic understanding can be tracked across layers

2 Testing Procedure

A full development of the testing procedure is available in the companion paper (Ashoff & Rodu, 2026) and an outline is provided here. The main test is a formal homogeneity test (Szekely & Rizzo, 2004) for embedding spaces based on energy statistics (You et al., 2022) and topological representations called persistence landscapes (Bubenik et al., 2015) and persistence silhouettes (Chazal et al., 2014). These topological methods result in stable representations of a model’s semantic structure that focus on the connectivity of the space rather than absolute scale, are invariant to homomorphic transformations, and are agnostic to the dimensionality of the embedding space. Regardless of the dimensionality these

techniques produce smooth two-dimensional curves. This allows the method to characterize the semantic structure or concept-space clustering more meaningfully than other graph matching or clustering methods.

Table 3 walks through the process of generating these representations from embeddings, running pairwise comparisons on these representations, and testing statistical significance at a Benjamini–Yekutieli (Benjamini & Yekutieli, 2001) adjusted level for multiple comparisons.

Input:	embeddings E_1, E_2 of the same inputs; bootstrap count n ; permutation count B ; significance level α
Output:	energy statistic T ; p-value p

- 1 for $r = 1, \dots, n$:
- 2 $S_1, S_2 \leftarrow \text{subsample}(E_1), \text{subsample}(E_2)$
- 3 $D_1, D_2 \leftarrow \text{persistence diagrams of } S_1, S_2$
- 4 transform D_1, D_2 (rotate and rescale; extra $H0$ rotation)
- 5 $\lambda_{1,r}, \lambda_{2,r} \leftarrow \text{landscape (or silhouette) of } D_1, D_2$
- 6 $T \leftarrow \text{energy statistic between } \{\lambda_1\} \text{ and } \{\lambda_2\}$
- 7 pool $\leftarrow \{\lambda_1\} \cup \{\lambda_2\}$
- 8 for $b = 1, \dots, B$:
- 9 randomly split pool into two sets of size n
- 10 $T^{(b)} \leftarrow \text{energy statistic between the permuted sets}$
- 11 $p \leftarrow (1 + \#\{b : T \leq T^{(b)}\}) / (B + 1)$
- 12 reject H_0 (equal semantic structure) at level α if $p \leq \alpha$

Table 3: *Topological alignment testing procedure*

Figure 1 shows three embeddings of 150 points each: two clean rings, three clean rings, and three noisy rings. Figure 2 shows the topological representation of these embeddings and the tests associated with these comparisons. The top two plots show the persistence landscapes and persistence silhouettes of the three embeddings. Especially in the landscapes, it is clear how the representations capture the clusters of points in the peaks of the plots. Three peaks for three clean rings, two peaks for two rings, and one peak as the noisy data blurs the underlying structure. The lines are shown with 95% pointwise confidence intervals.

The middle plot shows the test statistic when each embedding is compared to the two ring baseline. As expected, the test statistic is 0 and not significant when the two ring baseline is compared to itself. Then the clean three ring embedding is statistically different from the two rings, but more similar to the two ring embedding than the noisy three rings.

On the bottom are extra curve similarity metrics. The same topological representations are used and more standard similarity scores are shown to demonstrate the consistency of results. The Wasserstein distance is a natural metric for landscapes and the Jensen-Shannon Distance (JSD) is a natural metric for silhouettes, the scores are normalized within each representation. These metrics show similar values as the energy-statistics but prove to be less discriminative.

3 Concept Collapse via Language Model Adaptation

The first result demonstrates how this method can detect subtle failure modes. Increasingly adversarial LoRA adaptations are applied with the goal of degrading the model’s semantic structure, and the topological tests are able to track this degradation effectively.

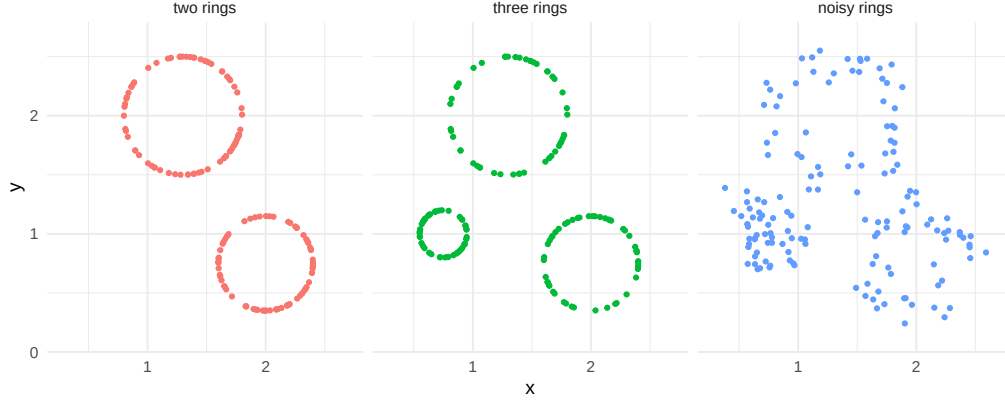


Figure 1: Three ring embeddings of 150 points each: two clean rings, three clean rings, and three noisy rings

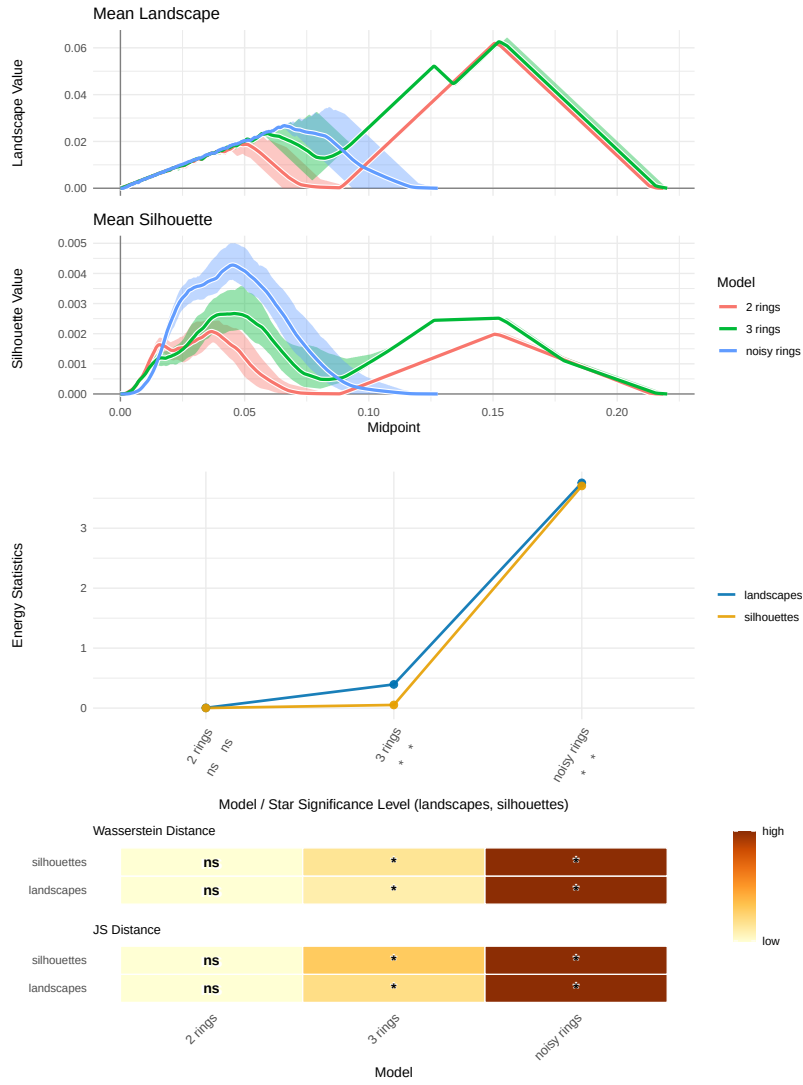


Figure 2: Ring Example Results: (top) Mean representations of three embeddings with 95% point-wise confidence bounds. (middle) Line plot of test statistics comparing each embedding to the two ring baseline with p -values represented in star notation under the model labels. (bottom) Heatmaps with the Wasserstein distance and JS distance between three embeddings and the two ring baseline.

Language models typically have vocabularies consisting of a collection of common words, subwords, single letters, and other prefixes or suffixes of words. This diversity of combinations allows the model to ingest any sequence of characters even if it is unfamiliar. How a model breaks apart words is called tokenization strategy and it varies significantly from model to model. However, if a word is subtokenized, or broken apart, it may be represented by multiple embeddings rather than a single vector. Many strategies aggregate multi-token sequences, but the different strategies may be difficult to compare. That question is avoided in this section, only examples where the word of interest (WOI) is not subtokenized are retained. This results in 358 samples for the OLMo model and 284 samples for the Llama model in the following results.

Similarly to how archetypes are defined for the image datasets of the companion paper (Ashoff & Rodu, 2026), archetype words are selected, and example usages of these words are identified. Using LLMs and these example sentences, the embeddings of a WOI is generated. Using the example sentences rather than the word alone allows the model to account for the context of the word.

Table 4 shows a sample of the word usages gathered from WordNet (Fellbaum, 1998), a “lexical database” that groups words into “cognitive synonyms (synsets)”. These synsets may not be the exact same word, but they are curated to represent similar meanings. The archetype words are chosen to be the words with the most synsets: break, cut, run, and play.

Word of Interest	Example Usage
break	he finally got his big break
fault	they built it right over a geological fault
broke	You broke the alarm clock when you took it apart!
cut	he played the first cut on the cd
track	the title track of the album
cut	She cut all of her major titles again
run	take a run into town
runs	the story or argument runs as follows
feeds	the Missouri feeds into the Mississippi
play	the play of light on the water
turn	it is my turn
play	the play of the imagination

Table 4: *Example words of interest and their usages*

To build a series of models, LoRAs are applied to a base model, Llama-3.2-1B available on Huggingface.² The development of the LoRAs uses the same $\frac{\alpha}{r}$ values as the LoRAs applied in the Art section, but only target Query and Value projections. However, instead

²huggingface.co/meta-llama/Llama-3.2-1B

of training the models to understand artistic style, these LoRAs try to convince the model that all the WOIs have the same definition. The LoRAs try to induce concept collapse, where each word loses its relationship to specific archetypes and each word becomes equally similar, or equidistant, to every other word. This collapse is accomplished through prompt engineering during LoRA training. Each example sentence for a given WOI is prepended by the same text:

"Learn the following definition. The phrase '<WOI>' means 'to go to an aquarium' in the context of the following sentence. <example sentence>"

Figure 3 shows that as increasingly aggressive LoRAs are applied, the models become more similar to the equidistant baseline. The downward trend shows that the concept collapse the LoRAs tried to introduce can successfully be tracked in the embedding space. Similarly, Figure 4 shows each of the LoRA models compared to the base model instead of the equidistant embedding. As increasingly aggressive LoRAs are applied, the effect can be tracked in the diverging semantic structure. This result is also the first time where the discriminative strength of the energy statistics is clear. While the scores of all three metrics show an upward swing, the Wasserstein and JS distances between the base model and the adapted model are only statistically significant for the more aggressive adaptations. The energy statistics on the other hand, retained significance in the more subtle adaptations.

4 Training Dynamics: Layer and Checkpoint Analysis

In addition to model selection and model adaptation, these tools can compare the internals of a model and the training process of a model. The Allen Institute for Artificial Intelligence (Ai2) develops open language models and releases the model training checkpoints alongside the final weights. These checkpoints allow for the investigation of the model’s semantic structure at each stage of training. The following results utilize the `OLMo-2-0425-1B` model from the OLMo 2 family available on Huggingface.³ OLMo 2 is a transformer style model with 4 trillion training tokens, 16 layers, and a 2048 hidden size. The same WordNet dataset is used as above, except no prompt engineering is used on the example sentences.

Figure 5 shows how the embedding space changes over the layers of a single model. The model has one input embedding and 16 layers. The plot shows the input and the first 15 layers compared to the final 16th layer. Especially towards the later layers, the embeddings change smoothly. Typically early layers are thought to encode coarse information while the later layers refine a model’s representation, consistent with the plot features: sharp changes of the early layers and then smooth refinement as it approaches the output layer. Similarly, Figure 6 shows the layers compared to an equidistant baseline. In the same way that the embedding space becomes more similar to the output layer deeper in the network, the embeddings also become more dissimilar to an equidistant embedding. This suggests that as the representations become more similar to the output space, the structure of the semantic space also increases.

Ai2 saves checkpoints of a model during training at least every 1000 training steps and publishes a total of 267 checkpoints for this model split into two main stages with the second stage further divided into three phases called “ingredients”. For the results, a sample of the full checkpoint list is used from across the training cycle: 15 checkpoints from stage 1 and 15 checkpoints from stage 2, 5 from each ingredient.

Stage 1, which the developers of OLMo (OLMo et al., 2024) describe as “pretraining”, is the longest stage of training. Figure 7 compares checkpoints to the untrained initialized model.

³huggingface.co/allenai/OLMo-2-0425-1B

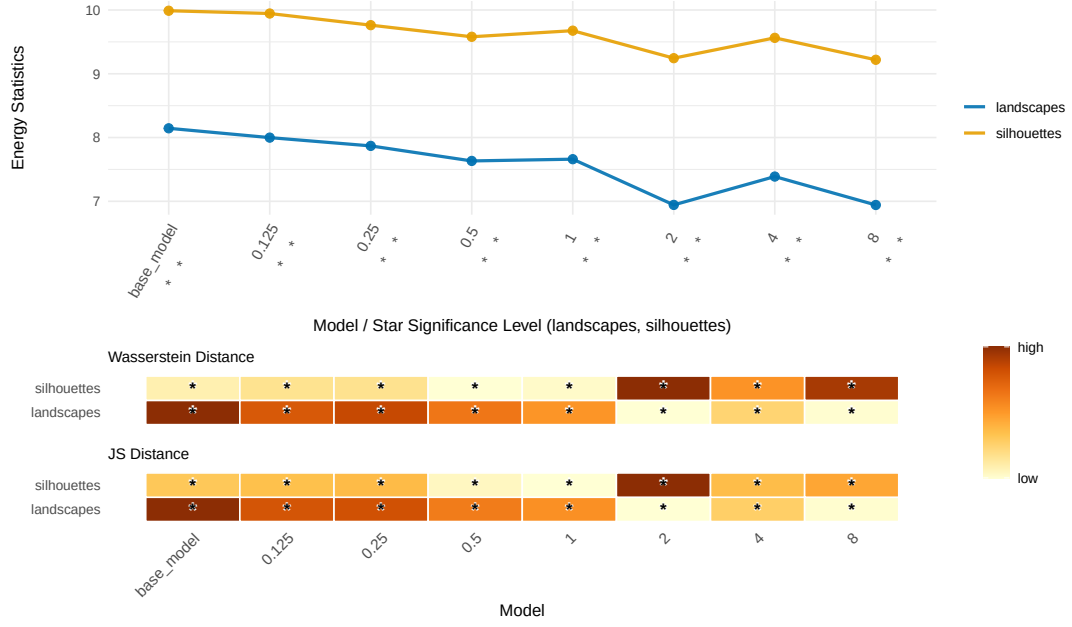


Figure 3: Comparisons of LoRA fine-tuned model embedding spaces to the equidistant baseline

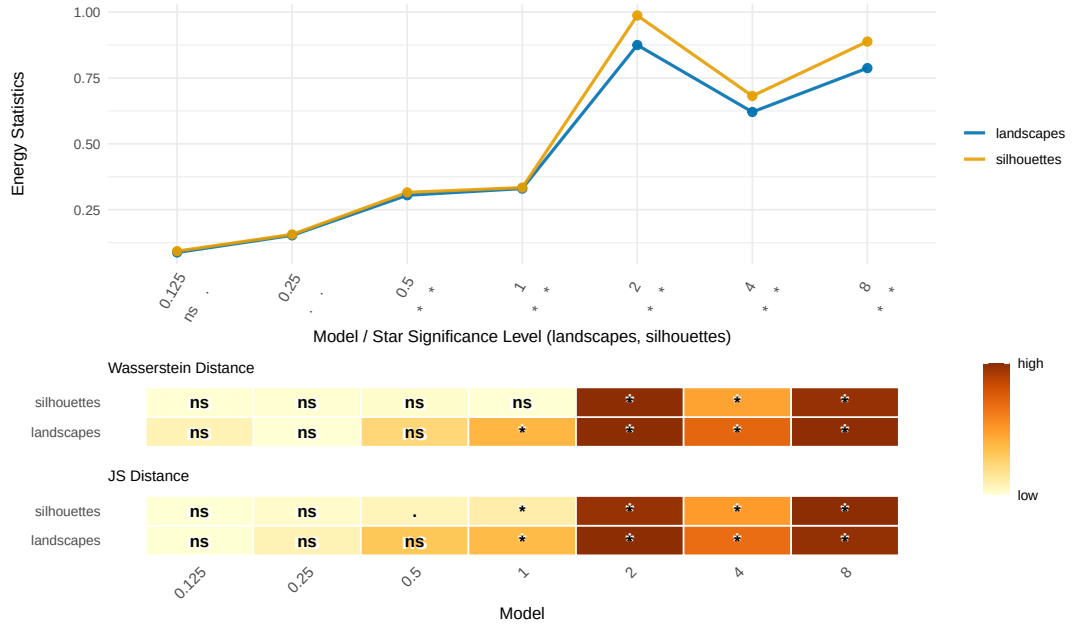


Figure 4: Comparisons of LoRA fine-tuned model embedding spaces to the base model

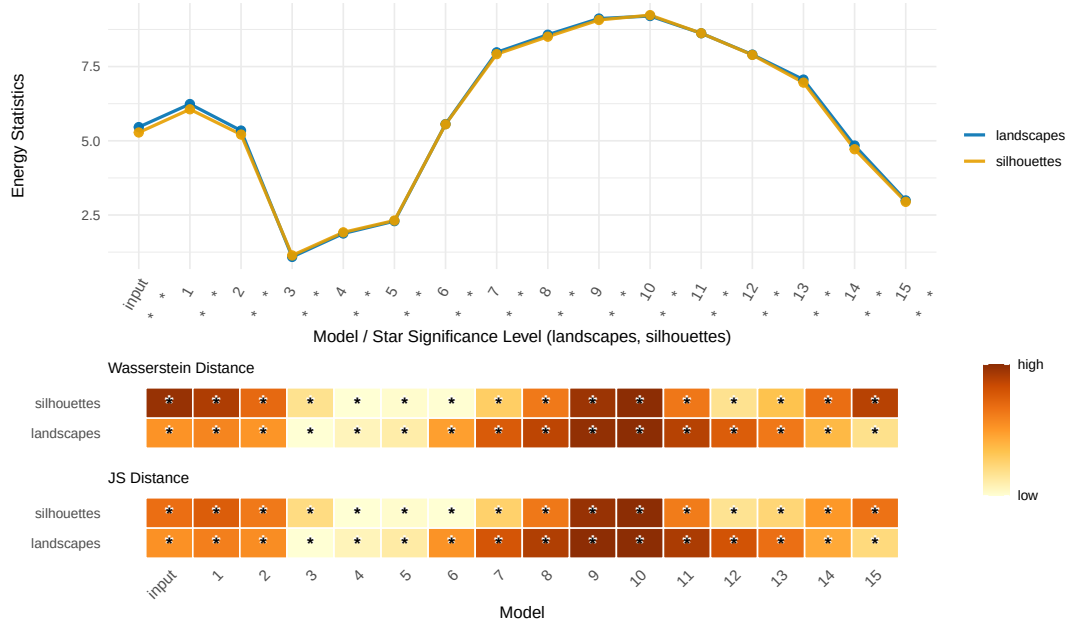


Figure 5: Comparisons of OLMo 2 layers to the output layer's embedding space

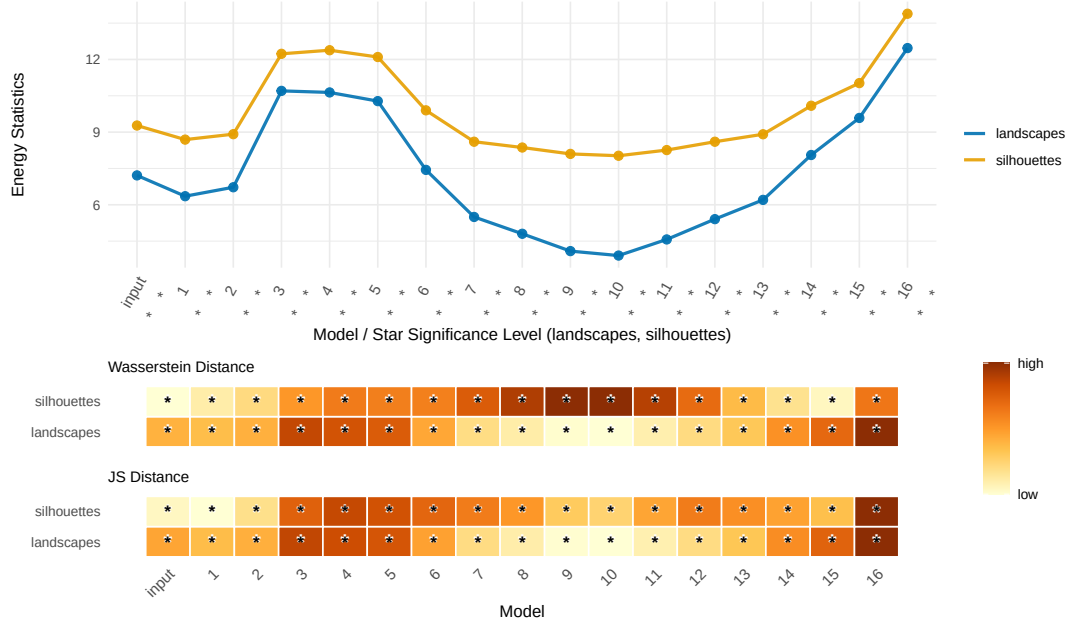


Figure 6: Comparisons of OLMo 2 layers to an equidistant baseline

A general trend exists, but the embeddings are inconsistent across checkpoints. This trend makes sense for a training stage designed for exploring a domain. Through the comparisons, changes in the embedding structure are tracked through the training process, but general pretraining does not have as clear of a signal as the more focused second stage.

Stage 2 is described as “mid-training”, where the model is tuned into a base model from which other fit-for-purpose models can be developed. During this phase, the learning rate is linearly decayed to 0 to stabilize the model. The same process is repeated three times to create three models, called “ingredients”. These models are averaged in a process the developers call “model souping” to create the final model.

Figures 8 and 9 compare checkpoints along the ingredients’ training process to the final checkpoint of stage 1 and the equidistant baseline respectively. Both comparisons show that while the learning rate is high, the models produce dissimilar semantic structures, but as the learning rate drops to 0, and the models stabilize, the structure converges. Similarly to the LoRA models, these tools give insight to the training processes.

5 Cross-Lingual Alignment: WMT14 Translation Models

Translating text is a common task for AI models. Having a source phrase in one language and a target phrase in another language (a translation pair) allows for an easy definition of a baseline. By separating the pairs by language, two datasets are created where the text of each entry is different but the meaning is the same. If a model is capturing the meaning of these phrases, then the embedding structure should look similar. Instead of comparing every model to a common baseline, for each model, the embedding of one language is compared to the other in the translation pair. This means that an energy statistic of 0 indicates exact structural alignment between both language embeddings.

Eight models are used to generate the embeddings. These models are all designed specifically for translation tasks, but they are developed with slightly different goals. Table 5 describes the main differences between the models⁴. Half the models are specifically designed for this type of embedding similarity task while the other half are generative models designed to output a translated text.

Unlike the previous language results, here the entire text is represented in a single embedding rather than isolating a single word. How these models encode the meaning of texts differs. Some models build up an understanding of the input as the input progresses so that the meaning of the text as a whole is captured in the last token. Others, especially the generative models, build up an encoding of the string that requires all the token embeddings to be averaged to recover the meaning of the whole text. Some models build in a “[CLS]” token to specifically hold the meaning of a text. The design choices affect the quality of the embedding alignment and this is reflected in the results.

⁴huggingface.co/facebook/bart-large-cnn, huggingface.co/google/translate-gemma-4b-it,
huggingface.co/Qwen/Qwen3-Embedding-8B, huggingface.co/google/madlad400-3b-mt,
huggingface.co/facebook/nllb-200-distilled-600m, huggingface.co/sentence-transformers/LaBSE,
huggingface.co/BAAI/bge-m3, huggingface.co/intfloat/multilingual-e5-large

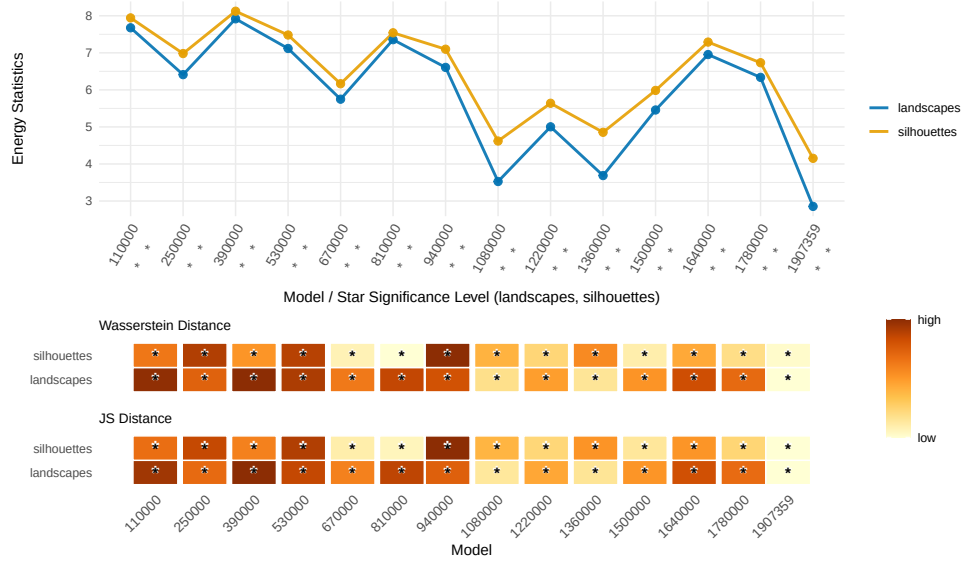


Figure 7: Comparisons of OLMo 2 pre-training checkpoints to the initialized model

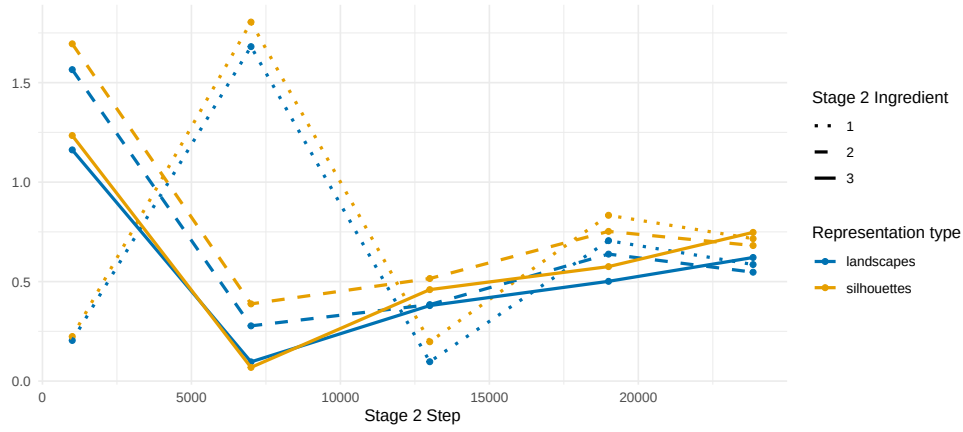


Figure 8: Comparisons of OLMo 2 mid-training "ingredient" checkpoints to the last pre-training checkpoint

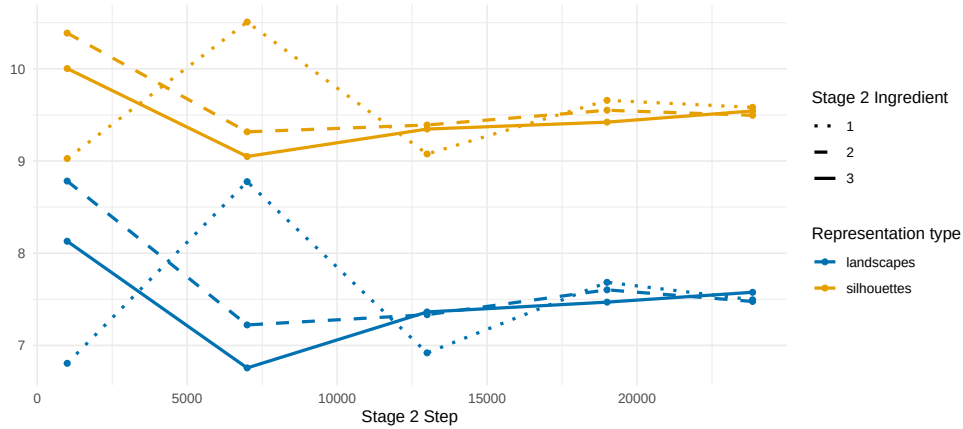


Figure 9: Comparisons of OLMo 2 mid-training "ingredient" checkpoints to the equidistant baseline

Model	Purpose	Readout.Strategy	Architecture
qwen3-embedding	Embedding	Last Token	Causal
LaBSE	Embedding	[CLS] Token	Bi-Directional
BGE-M3	Embedding	[CLS] Token	Bi-Directional
multilingual-e5-large	Embedding	Mean Pooling	Bi-Directional
bart-large-cnn	Generative	Mean Pooling	Encoder-Decoder
translategemma-4b-it	Generative	Last Token	Causal
madlad400-3b-mt	Generative	Mean Pooling	T5
nllb-200-distilled-600m	Generative	Mean Pooling	Encoder-Decoder

Table 5: Translation models’ architectures and readout strategies: (Lewis et al., 2019), (Finkelstein et al., 2026), (Qwen Team, 2025), (Kudugunta et al., 2023), (Costa-Jussà et al., 2022), (Feng et al., 2022), (Chen et al., 2024), (Wang et al., 2024)

Figures 10, 11, 12 show the results of comparing the embeddings generated by all the models with 200 translation pairs each for three language pairs: French-English, German-English, and Hindi-English. These language pairs are pulled from the Workshop on Machine Translation 2014 (WMT14) (Bojar et al., 2014) dataset available on Huggingface.⁵ This is a relatively old dataset and many of the new models will likely have ingested the language pairs during training, but it is a well studied and classic dataset.

In the graphs, the models are ordered by the average of their landscape and silhouette energy statistics such that the models with the most aligned embedding spaces are towards the right. Across the three language pairs, the embedding models generally out perform the generative models on alignment. This is expected because they are trained specifically for embedding alignment tasks. If a general model is needed, the BGE-M3 embedding model may be the best choice, but the generative MADLAD model may be better suited to the Hindi-English translations. Similarly, the Qwen3-Embedding model appears to excel at German-English alignment but may fall behind even generative models in French-English and Hindi-English alignment.

⁵huggingface.co/datasets/wmt/wmt14

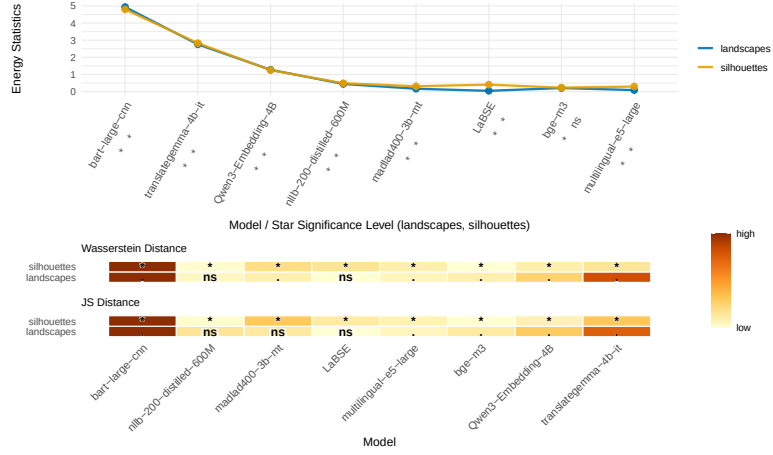


Figure 10: Comparing embedding spaces of French and English translation pairs across translation models

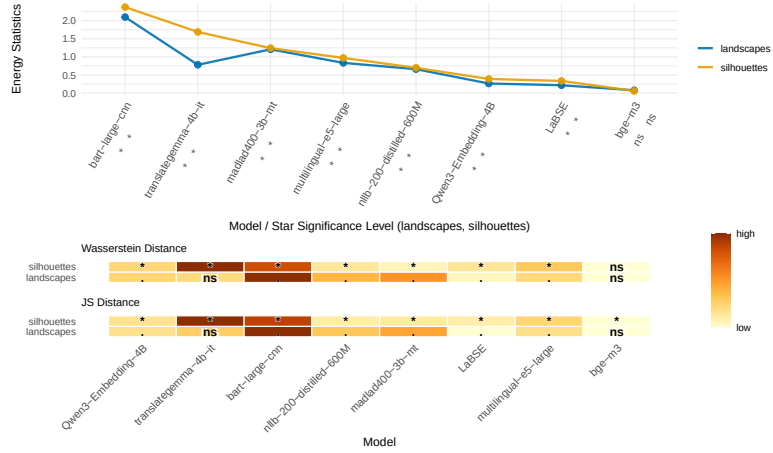


Figure 11: Comparing embedding spaces of German and English translation pairs across translation models

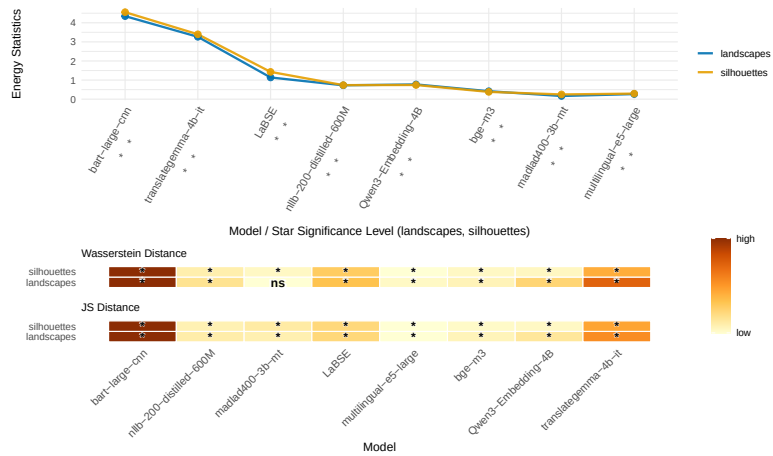


Figure 12: Comparing embedding spaces of Hindi and English translation pairs across translation models

6 Discussion and Conclusion

These results show that this topological method can be used to investigate the semantic space of LLMs. The first result demonstrates that the technique can identify and track the manipulation of model behavior. By applying increasingly aggressive adaptations to the model this method tracks the changes in the semantic space. The baseline selection is important in this result. By comparing against an equidistant baseline and against a base model, this result shows that comparisons to model embedding spaces and human knowledge structures can be made rigorously under a single framework.

The next result, tracking the internal representation of a model over layers and over training cycles is closer to a discovery result. Rather than comparing to an external baseline, the comparisons track how the semantic structure changes as an output is produced. Anthropomorphically, this lets the thought process of the model become visible.

The final result is a demonstration of model selection by testing textual meaning. Across multiple language pairs, the similarity of meaning is tested. Since the input texts are direct translations, the semantic structure should be exactly the same. This exact matching is not so different from exact string matching, but the extensions are much broader. This result demonstrates capability on tightly matching texts, but it can be just as easily used to match complex translations of long texts or idiomatic language which is difficult to translate verbatim, but may have a strong semantic match.

Together these results show how model selection, behavior manipulation, and discovery of internal mechanisms can all be accomplished under a single alignment framework. This multi-modal and dimension-agnostic technique does not replace outcome reasoning, it fills in gaps in understanding left open by other evaluation methods and aligns model developments with curated semantic preferences.

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